OAK INTERNATIONAL SCHOOL

EXCELLENCE ENSEMBLE • DUAL CURRICULUM MATHEMATICS

IB MIDDLE YEARS & DIPLOMA	BENIN NATIONALE APC	BILINGUAL EN / FR

PREMIUM MATHEMATICS ACTIVITY WORKBOOK

OFFICIAL CURRICULUM RESEARCH REPORT

A Harmonised Dual-Curriculum Mathematics Learning Companion

Connecting Conceptual Inquiry, Axiomatic Proof, and Examination Mastery

Target Cohort:	Official Curriculum Research Report
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Comprehensive Mathematics Curriculum Research Report: IB Middle Years & Diploma Programmes and the Beninese Approche Par Compétences (APC)

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EXECUTIVE SUMMARY

Oak International School delivers a dual-curriculum educational model in which students concurrently prepare for:

1. The **International Baccalaureate (IB)** continuum: Middle Years Programme (MYP 1–5) and Diploma Programme (DP 1–2), taught in English;
2. The **Beninese National Mathematics Curriculum**, structured under the **Approche Par Compétences (APC)** and implemented in French across Primary (CM2), Lower Secondary (Collège: 6ème–3ème), and Upper Secondary (Lycée: 2nde–1ère). The Terminale year is explicitly excluded from the workbook series but forms the terminal horizon of the national cycle.

This research report provides the empirical and documentary foundation for pedagogical harmonisation. It establishes:

- The precise statutory mandates, competency taxonomies, and examination frameworks of both systems;
- The assessment paradigms: IB Criteria A–D and DP Papers 1–3 / Internal Assessment versus Beninese criteria: *Analyser* (C_a), *Mathématiser* (C_m), *Opérer* (C_o), and *Critère de Perfectionnement* (CP);
- An evidential inventory of official sources, categorised by institutional reliability;
- A rigorous feasibility evaluation of concurrent delivery, defining where deep synergy is achievable and where divergent curricular structures require targeted instructional strategies.

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1. OFFICIAL CURRICULUM FRAMEWORK: INTERNATIONAL BACCALAUREATE (IB)

1.1 Middle Years Programme (MYP) Mathematics (MYP 1 to MYP 5)

The IB MYP mathematics framework spans ages 11 to 16. In Oak International School's grade mapping:

- **MYP 1** corresponds to **CM2** (Grade 5 / Class 6)
- **MYP 2** corresponds to **Sixième (6ème)**
- **MYP 3** corresponds to **Cinquième (5ème)**
- **MYP 4** corresponds to **Quatrième (4ème)**
- **MYP 5** corresponds to **Troisième (3ème)**

1.1.1 Curricular Architecture and Content Branches

The MYP Mathematics framework is structured into four universal content branches:

1. **Number**: Forms of number, scientific notation, estimation, sets, rational and irrational numbers, number theory, surds, exponents, and proportional reasoning.
2. **Algebra**: Algebraic manipulation, factorisation, expansion, linear and quadratic equations/inequalities, systems of equations, sequences and series, functions and graphs, and mathematical modelling.
3. **Geometry and Trigonometry**: Angle relationships, circle geometry, congruence and similarity, 2D and 3D mensuration (perimeters, areas, surface areas, volumes), coordinate geometry, Pythagoras' theorem, right-angled trigonometry, and vectors and transformations.
4. **Statistics and Probability**: Data collection and sampling, measures of central tendency and dispersion, frequency distributions, cumulative frequency, graphical representations (box-and-whisker plots, histograms, scatter plots), bivariate data and line of best fit, theoretical and experimental probability, compound events, tree diagrams, and Venn diagrams.

In MYP 4 and 5, schools may provide **Standard Mathematics** (broad foundation) and **Extended Mathematics** (including topics such as radical equations, matrices, fractional exponents, advanced trigonometry with sine/cosine rules, and introductory differentiation).

1.1.2 Conceptual Lens: Key and Related Concepts

- **Key Concepts**: *Form*, *Logic*, and *Relationships*.
- **Related Concepts**: *Change*, *Equivalence*, *Generalization*, *Justification*, *Measurement*, *Models*, *Pattern*, *Quantity*, *Representation*, *Space*, *System*, and *Validity*.
- **Global Contexts**: Contextualisation through lenses such as *Scientific and technical innovation*, *Fairness and development*, and *Globalization and sustainability*.

1.1.3 Assessment Taxonomy (MYP Criteria A–D)

Summative achievement in MYP Mathematics is evaluated against four criteria, each scored on an 8-level achievement band (0–8, maximum score of 32):

- **Criterion A: Knowing and understanding**
- Select appropriate mathematics when solving problems in familiar and unfamiliar situations.

- Apply the selected mathematics successfully when solving problems.
- Solve problems correctly in a variety of contexts.
- **Criterion B: Investigating patterns**
- Select and apply mathematical problem-solving techniques to discover complex patterns.
- Describe patterns as general rules consistent with correct findings.
- Prove, or verify and justify, these general rules.
- **Criterion C: Communicating**
- Use appropriate forms of mathematical representation to present information.
- Move between different forms of mathematical representation.
- Communicate complete, coherent, and concise lines of mathematical reasoning.
- Organize information using a logical structure.
- **Criterion D: Applying mathematics in real-life contexts**
- Identify the relevant elements of authentic real-life situations.
- Select appropriate mathematical strategies when solving authentic real-life situations.
- Apply the selected mathematical strategies successfully to reach a valid solution.
- Explain the degree of accuracy of each solution and whether the solution makes sense in the real-life context.

1.2 Diploma Programme (DP) Mathematics (DP 1 and DP 2)

In the Oak structure:

- **DP 1** corresponds to **Seconde (2nde)**
- **DP 2** corresponds to **Première (1ère)**

1.2.1 Course Pathways and Levels

Since the September 2019 curriculum reform (first examination May 2021), DP Mathematics offers two distinct pathways:

1. Mathematics: Analysis and Approaches (AA):

- Designed for students fascinated by pure mathematics, deductive reasoning, abstract algebraic structures, rigorous formal proof, and analytical calculus.
- Available at **Standard Level (SL)** (150 hours) and **Higher Level (HL)** (240 hours).

2. Mathematics: Applications and Interpretation (AI):

- Designed for students oriented toward applied mathematics, statistical inference, mathematical modelling, data analysis, dynamic systems, and technological computation.
- Available at **Standard Level (SL)** (150 hours) and **Higher Level (HL)** (240 hours).

1.2.2 The 60-Hour Common Core Framework

All four DP courses share a 60-hour common core comprising five syllabus areas:

- **Topic 1: Number and Algebra**: Scientific notation, arithmetic and geometric sequences/series, financial mathematics (compound interest, annual percentage rates), laws of logarithms, proof techniques.
- **Topic 2: Functions**: Function concept, domain, range, inverse functions, composite functions, quadratic models, rational functions, exponential and logarithmic models.
- **Topic 3: Geometry and Trigonometry**: 3D distance and angles, radian measure, trigonometric ratios, unit circle, sine and cosine rules, area of triangles, sectors, arcs.
- **Topic 4: Statistics and Probability**: Sampling methods, frequency polygons, boxplots, variance and standard deviation, correlation (r) and linear regression, probability rules, Venn diagrams, discrete random variables, normal distribution.
- **Topic 5: Calculus**: Limits, first principles, power rule, derivatives of polynomials, equation of tangent and normal, optimization (extrema), indefinite and definite integration, area under curves.

1.2.3 Course Differentiation: AA versus AI, SL versus HL

- **Analysis and Approaches (AA SL/HL)** emphasizes:
- Deep algebraic manipulation, analytic trigonometry (sin(A± B), double angles, trigonometric proof).
- Advanced calculus (chain rule, product/quotient rules, related rates, integration by substitution, integration by parts, differential equations).
- HL extensions: mathematical induction, contradiction, complex numbers (z = a+ib = re^iθ), De Moivre's theorem, vector equations of lines and planes in 3D, Taylor/Maclaurin series, L'Hôpital's rule.
- Non-calculator requirement in Paper 1.
- **Applications and Interpretation (AI SL/HL)** emphasizes:
- Pervasive technology use (Graphic Display Calculator - GDC).
- Advanced statistical inference: Chi-squared tests (chi² independence and goodness of fit), t-tests, Poisson and binomial distributions, confidence intervals.
- Mathematical modelling: logistic, sinusoidal, power models.
- HL extensions: matrices and Markov chains, graph theory (algorithms: Dijkstra, Kruskal, Chinese Postman, Travelling Salesperson), coupled linear differential equations, phase portraits.

1.2.4 Assessment Structure

- **DP External Assessment**:
- **AA SL**: Paper 1 (90 min, 80 marks, **No Calculator**); Paper 2 (90 min, 80 marks, **GDC Required**).
- **AA HL**: Paper 1 (120 min, 110 marks, **No Calculator**); Paper 2 (120 min, 110 marks, **GDC Required**); Paper 3 (60 min, 55 marks, extended investigative problem, **GDC Required**).
- **AI SL**: Paper 1 (90 min, 80 marks, short questions, **GDC Required**); Paper 2 (90 min, 80 marks, extended questions, **GDC Required**).
- **AI HL**: Paper 1 (120 min, 110 marks, **GDC Required**); Paper 2 (120 min, 110 marks, **GDC Required**); Paper 3 (60 min, 55 marks, two extended modelling tasks, **GDC Required**).
- **Internal Assessment (IA)** (20% of final grade across all courses):
- Individual Mathematical Exploration (12–20 pages), assessed on Criterion A (Presentation), Criterion B (Mathematical communication), Criterion C (Personal engagement), Criterion D (Reflection), and Criterion E (Use of mathematics).

1.2.5 Determination of Oak International School's DP Pathway

Because Oak operates in Benin, where the national upper-secondary curriculum is strongly algebraic, vector-based, and analytical (Series C and D), the most compatible pathway is **Mathematics: Analysis and Approaches (AA)**. However, where students pursue social sciences, **Mathematics: Applications and Interpretation (AI)** may be offered. Therefore, this project provides a **provisional common-core harmonisation framework** combined with modular pathway-specific extensions.

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2. OFFICIAL CURRICULUM FRAMEWORK: BENINESE APPROCHE PAR COMPÉTENCES (APC)

2.1 Governance and Institutional Architecture

The Beninese mathematics curriculum is established under statutory orders of:

1. **Ministère des Enseignements Maternel et Primaire (MEMP)**: Direction de l'Enseignement Primaire (DEP) and Institut National pour la Formation et la Recherche en Éducation (INFRE) for Primary (CI to CM2).
2. **Ministère de l'Enseignement Secondaire, Technique et de la Formation Professionnelle (MESTFP)**: Direction de l'Inspection Pédagogique, de l'Innovation et de la Qualité (DIPIQ) and Direction des Examens et Concours (DEC) for Secondary (6ème to Terminale).

The secondary mathematics curriculum derives from the regional *Harmonisation des Programmes de Mathématiques (HPM)* developed across French-speaking Africa under the auspices of the Conférence des Ministres de l'Éducation des pays ayant le français en partage (CONFEMEN) and operationalised in the *Collection Inter-Africaine de Mathématiques (CIAM)* textbooks and national pedagogical guides.

2.2 Primary Mathematics Curriculum (CM2)

The CM2 curriculum is structured around the development of terminal capacities evaluated at the **Certificat d'Études Primaires (CEP)**:

- **Arithmétique et Nombres**: Whole numbers up to billions, decimal numbers, operational mechanics (addition, subtraction, multiplication, long division with decimal quotients), fractions (simplification, equivalence, addition/subtraction with like/unlike denominators, fraction of an amount).
- **Grandeurs et Mesures**: Metric system (length: km to mm; mass: t, q, kg, g; capacity: kL to mL; area: m^2, dm^2, cm^2, and agrarian units: hectare, are, centiare; volume: m^3, dm^3, cm^3 and the fundamental equivalence 1 dm^3 = 1 L). Time and duration: hours, minutes, seconds, operations on compound units, speed ($d = v \times t$).
- **Système Financier et Vie Économique**: Commercial arithmetic: purchase price (*prix d'achat*), transport/overhead expenses (*frais*), cost price (*prix de revient*), selling price (*prix de vente*), profit (*bénéfice*), loss (*perte*), percentage discount and markup, simple interest ($I = (C \times t \times n / 100)$), capital, investment rate.
- **Géométrie**: Perpendicular and parallel lines, angles (droit, aigu, obtus, plat), triangles (classification by sides and angles, constructions), quadrilaterals (square, rectangle, rhombus, parallelogram, trapezium: characteristics and perimeter/area formulas), circle and disc (radius, diameter, circumference $P = 2\pi r$, area $A = \pi r^2$), geometric solids (cube, cuboid, cylinder: net/patron and volume calculation).

2.3 Lower Secondary Mathematics Curriculum (Premier Cycle: 6ème to 3ème)

The lower-secondary curriculum is structured under the **Nouveaux Programmes d'Études (NPE)** according to the APC.

2.3.1 Competency Architecture

The Beninese curriculum articulates three **Compétences Disciplinaires (CD)**:

- **CD 1**: *Résoudre des situations-problèmes en utilisant les configurations géométriques de l'espace et du plan ainsi que les transformations du plan.* (Spatial, geometric, and transformational competence).
- **CD 2**: *Résoudre des situations-problèmes en utilisant les nombres, le calcul littéral et l'organisation des données.* (Numerical, algebraic, and data competence).
- **CD 3**: *Appréhender la mathématique dans ses aspects numériques et graphiques par l'appropriation des outils, techniques et procédés conventionnels ainsi que par le traitement de l'information.* (Modelling, procedural tools, and communication).

Cross-curricular competence is developed through:

- **Compétences Transversales (CTV)**: Exploitation of information, problem solving, critical thinking, creative communication, cooperative work.
- **Compétences Transdisciplinaires (CT)**: Integration of academic learning into socio-economic, environmental, and civic life.

2.3.2 Pedagogical Structure: Situations d'Apprentissage (SA)

Each academic year is organized into canonical *Situations d'Apprentissage* (Learning Situations), each comprising:

1. *Situation de départ* (Contextual problem statement).
2. *Cheminement de l'apprenant* (Learner trajectory: appropriation of the context, formulation of mathematical questions, experimentation/investigation, acquisition of new knowledge, decontextualisation, deepening activities, transfer, and evaluation).

2.3.3 Grade-by-Grade Content Breakdown

- **Sixième (6ème)**:
- *SA 1: Configurations du plan*: Points, straight lines, segments, rays, intersections, parallelism, perpendicularity (construction with compass and square), midpoint, circle, angles (measurement with protractor, types), perpendicular bisector (*médiatrice*).
- *SA 2: Nombres et calculs*: Natural numbers (**N**), arithmetic decimals (**D**^+), division algorithm with remainder, divisibility rules (2, 3, 5, 9, 10), fraction concept, comparison of fractions, rounding (*arrondi*) and truncation (*troncature*).
- *SA 3: Configurations de l'espace*: Rectangular cuboid (*pavé droit*), cube, right prism (*prisme droit*): cavalier perspective, nets, surface areas, volumes.
- *SA 4: Organisation des données et grandeurs*: Metric conversions, perimeter and area of basic shapes, direct proportionality tables, scale, percentages, bar charts.
- *SA 5: Applications du plan*: Axial symmetry (*symétrie orthogonale par rapport à une droite*), construction of symmetric points, conservation of lengths, angles, alignments, and areas.
- **Cinquième (5ème)**:
- *SA 1: Configurations de l'espace*: Right circular cylinder, right prism, right cone, sphere: identification, nets, volume and surface area.
- *SA 2: Configurations du plan*: Angles formed by two parallel lines and a secant (alternate interior, alternate exterior, corresponding), vertically opposite angles; sum of angles in a triangle (180^circ); triangle inequality; special lines in triangles (mediators/circumcentre, angle bisectors/incentre); parallelograms and special quadrilaterals.
- *SA 3: Nombres et calculs*: Relative integers (**Z**) and relative decimals: ordering, number line, addition and subtraction of signed numbers, removal of parentheses; fraction operations (addition, subtraction with common or multiple denominators); positive integer powers of 10.
- *SA 4: Applications du plan*: Central symmetry (*symétrie centrale* / point reflection), centre of symmetry, properties of invariance, characterisation of parallelograms by central symmetry.
- *SA 5: Organisation des données*: Proportionality factor, speed/distance/time, scale drawings, statistical frequency tables, bar and pie charts.
- **Quatrième (4ème)**:
- *SA 1: Configurations du plan*: Right-angled triangle and circumscribed circle; Pythagoras' Theorem and its converse; cosine of an acute angle in a right-angled triangle; midpoint theorem in triangles (*théorème des milieux*); concurrency of special lines in triangles (medians/centroid G, altitudes/orthocentre H).
- *SA 2: Nombres et calculs*: Rational numbers (**Q**): operations (+, -, ×, ÷), sign rules, irreducible fractions; integer exponents (positive and negative powers of numbers); square roots of perfect squares; literal arithmetic: algebraic expressions, expansion, factorisation using single distributivity, introduction to linear equations (ax + b = 0).
- *SA 3: Configurations de l'espace*: Regular pyramids and right cones: perspective representation, lateral and total area, volume.
- *SA 4: Applications du plan*: Translations and introduction to vectors: definition of vector by direction, sense, and norm; equality of vectors, Chasles' relation (AB■ + BC■ = AC■).
- *SA 5: Organisation des données*: Cumulative frequencies, weighted arithmetic mean, median, linear proportional relationships (f(x) = ax).
- **Troisième (3ème)**:
- *SA 1: Configurations du plan*: Thales' Theorem and its converse in triangles; trigonometry in right triangles: sine, cosine, tangent, fundamental trigonometric identities ($\sin^2 x + \cos^2 x = 1$, $\tan x = (\sin x / \cos x)$); inscribed angles and central angles in a circle; regular polygons.
- *SA 2: Nombres et calculs littéraux*: Real numbers (**R**); radicals/square roots: algebraic manipulation, simplification $\sqrt{(a^2 b)} = a\sqrt{(b)}$, rationalisation of denominators; algebraic identities: $(a+b)^2$, $(a-b)^2$, $(a-b)(a+b)$; complete factorisation of polynomials; rational fractions (domain of validity, simplification); linear equations and inequalities, null-factor equations (ax+b)(cx+d)=0; linear systems of two equations in two unknowns (2 × 2).
- *SA 3: Géométrie analytique et vecteurs*: Orthonormal coordinate system (O, i■, j■); vector coordinates u■(x, y), midpoint coordinates, distance formula $d = \sqrt{((x_B-x_A)^2 + (y_B-y_A)^2)}$; collinearity condition (xy' - x'y = 0); orthogonality condition (xx' + yy' = 0); equations of straight lines (y = mx + p or ax + by + c = 0), slopes of parallel and perpendicular lines.
- *SA 4: Configurations de l'espace*: Truncated cones and pyramids, plane sections of 3D solids, scaling factor impacts on lengths (k), areas (k^2), and volumes (k^3).

- *SA 5: Organisation des données et fonctions*: Affine functions (f(x) = ax + b), slope and y-intercept, graphical interpretation; statistics of grouped continuous data: class intervals, histograms, modal class, mean calculation.

2.3.4 National Examination: Brevet d'Études du Premier Cycle (BEPC)

The BEPC examination in Benin evaluates mathematical competence through a unified, contextualised *Situation d'Évaluation* featuring:

- A realistic context (e.g., regional development, cooperative farming, logistics);
- An overarching task (*Tâche*);
- Three mandatory structured sub-problems (*Problèmes 1, 2, et 3*).

Evaluation is executed according to three criterion rubrics:

1. **Critère 1 : Analyser (C_a)**: Relevance of interpretation, extraction of given data, identification of unknown quantities, and formulation of mathematical hypotheses.
2. **Critère 2 : Mathématiser (C_m)**: Correct selection of theorems, algebraic equations, geometric properties, or statistical formulas.
3. **Critère 3 : Opérer (C_o)**: Precision and correctness of calculations, algebraic transformations, geometric constructions, and coherence of deductive chains.
4. **Critère de Perfectionnement (CP)**: Presentation, mathematical notation, handwriting, and layout (typically 2 points out of 20).

2.4 Upper Secondary Mathematics Curriculum (Second Cycle: 2nde and 1ère)

In Benin, the upper-secondary cycle is tracked into streams. For Oak International School students preparing for international university entrance and the IB Diploma, the relevant Beninese programmes are **Série C** (Mathematics & Physical Sciences) and **Série D** (Life/Earth Sciences & Mathematics).

2.4.1 Seconde (2nde C / D)

- **Calcul numérique et algébrique dans R**: Subsets of **R** (**N** ⊂ **Z** ⊂ **D** ⊂ **Q** ⊂ **R**), intervals, absolute value (|x-a| ≤ r), bounds (majorant, minorant), order in **R**; polynomial functions, roots, factorisation, quadratic equations and inequalities using the discriminant Δ = b² - 4ac and sign tables.
- **Fonctions numériques**: General concepts of mapping and function, domain of definition, parity (even/odd), periodicity, monotonicity, extrema, reference functions (x mapsto x², x mapsto (1 / x), x mapsto √(x), x mapsto |x|, affine functions), composite transformations (f(x) + k, f(x+k), -f(x)).
- **Géométrie vectorielle et produit scalaire**: Vector spaces in dimension 2, bases and coordinates, dot product of two vectors (u■ cdot v■ = |u■| |v■| cos θ = xx' + yy'), orthogonal projections, Al-Kashi theorem (Law of Cosines: a² = b² + c² - 2bc cos hatA), median theorem, cartesian equations of circles.
- **Trigonométrie**: The unit circle, radian measure, trigonometric lines (cos x, sin x, tan x), addition formulas (cos(a ± b), sin(a ± b)), double-angle formulas (cos 2a, sin 2a).
- **Géométrie de l'espace**: Incidence axioms, lines and planes in 3D, relative positions, criteria for parallelism and orthogonality between lines and planes.
- **Statistiques à une variable**: Grouped data, parameters of position (mean, median, quartiles), parameters of dispersion (variance, standard deviation, interquartile range), box plots.

2.4.2 Première (1ère C / D)

- **Barycentres de points pondérés**: Weighted points (A, α), condition sum α_i ≠ 0, definition sum α_i GA_i■ = 0■, coordinates of barycentres, partial barycentres theorem (associativity), application to geometric alignment and concurrency proofs, level curves (*lignes de niveau*: e.g., |aMA■ + bMB■| = k and MA■ cdot MB■ = k).
- **Compléments sur les polynômes et équations**: Second-degree equations with parameters, sign of roots, bi-quadratic equations, systems of non-linear equations, linear programming via half-planes.
- **Dérivation et étude de fonctions**: Derivative at a point (limit of difference quotient), geometric interpretation as tangent slope, derivative functions of power, rational, and radical functions, differentiation rules (sum, product, quotient), relationship between sign of derivative and monotonicity, local extrema, asymptotes (vertical, horizontal, oblique: lim_x→∈fty [f(x) - (ax+b)] = 0), curve sketching.
- **Suites numériques**: Explicit and recursive sequences, arithmetic and geometric sequences (general term, sum of first n terms), monotonicity, boundedness, intuitive convergence.
- **Trigonométrie avancée**: Equations cos x = cos α, sin x = sin α, tan x = tan α, product-to-sum and sum-to-product transformation formulas.
- **Dénombrement et Probabilités**: Combinatorics: Cartesian product, permutations, arrangements (A_n^p), combinations (C_n^p or binomnp), Pascal's triangle, binomial theorem; classical probability on finite sample spaces, conditional probability, independent events, discrete random variables, probability distributions, expectation E(X), variance V(X), standard deviation σ(X).
- **Transformations du plan et géométrie de l'espace**: Homotheties, rotations, planar isometries; orthogonality and vector dot products in 3D space.

2.4.3 Terminal Horizon: Baccalauréat (Séries C et D)

The Beninese Baccalauréat examination in mathematics requires 4 hours (coefficient 6 in Série C, coefficient 4 in Série D) and comprises three extensive contextualised problems requiring rigorous deductive proofs, curve sketching, parametric calculus, spatial geometry, and probability distributions.

3. RESEARCH PROTOCOL AND SOURCE INVENTORY

To ensure complete evidential integrity, all curricular assertions in this study are grounded in statutory publications and established institutional syllabi, verified across three tiers of reliability:

- **Tier 1 (Authoritative Official)**: Statutory curriculum guides and examination specifications published directly by the International Baccalaureate Organization (IBO), the Ministère des Enseignements Maternel et Primaire (MEMP), or the Ministère de l'Enseignement Secondaire, Technique et de la Formation Professionnelle (MESTFP).
- **Tier 2 (Official Regional / Examination Archives)**: Regional harmonised curricular frameworks (CONFEMEN/HPM) and official national examination papers/scoring rubrics (CEP, BEPC, and Baccalauréat Bénin).
- **Tier 3 (Corroborating Pedagogical Standard)**: Established African pedagogical textbooks adopted by the Beninese national education system (*Collection Inter-Africaine de Mathématiques - CIAM*, EDICEF/CEDA).

Ref ID	Curriculum & Level	Mathematical Domain / Competency	Verified Learning Expectation / Prescribed Content	Document Reference & Publishing Institution	Source Reliability	Research Notes / Status
SRC-01	IB MYP (Years 1–5)	All Branches / Assessment	Assessment Criteria A, B, C, D (8 bands each); Conceptual framework (Form, Logic, Relationships)	*Middle Years Programme Mathematics Guide*, International Baccalaureate Organization (Cardiff/Geneva, 2020)	Tier 1 (Official IB)	Primary statutory guide for MYP mathematics continuum.
SRC-02	IB DP (Years 1–2)	Analysis & Approaches / Applications & Interpretation	60-hour Common Core; SL/HL differentiation; Paper 1 (non-calc in AA), Paper 2, Paper 3 (HL), Internal Assessment Exploration	*Diploma Programme Mathematics: Analysis and Approaches Guide & Applications and Interpretation Guide*, IBO (Geneva, 2019/2021)	Tier 1 (Official IB)	Statutory specification of current DP pathways.
SRC-03	Benin CM2	Nombres, Calcul, Mesures, Géométrie	Four operations, fractions, metric conversions, perimeter/area/volume, commercial calculations, speed	*Programme d'études de mathématiques au Cours Moyen Deuxième Année (CM2)*, MEMP / Direction de l'Enseignement Primaire (Porto-Novo, Benin)	Tier 1 (Official Ministry)	Foundation for Certificat d'Études Primaires (CEP).
SRC-04	Benin 6ème	Configurations du plan & espace, Nombres	Point, droite, cercle, médiatrice, symétrie axiale; **N**, **D**^+, fractions simples; pavé droit, cube	*Guide pédagogique de mathématiques 6ème (Nouveaux Programmes d'Études)*, MESTFP / DIPIQ (Cotonou, Benin)	Tier 1 (Official Ministry)	Initial secondary APC implementation.
SRC-05	Benin 5ème	Angles, symétrie centrale, **Z**, solides	Angles alternes-internes; somme des angles d'un triangle; symétrie centrale; entiers relatifs **Z**; cylindre, cône	*Guide pédagogique de mathématiques 5ème (NPE)*, MESTFP / DIPIQ (Cotonou, Benin)	Tier 1 (Official Ministry)	Corroborated with CIAM 5ème.
SRC-06	Benin 4ème	Pythagore, rationnels, vecteurs, statistiques	Théorème de Pythagore; nombres rationnels **Q**; puissances entières; calcul littéral; vecteurs et translation	*Guide pédagogique de mathématiques 4ème (NPE)*, MESTFP / DIPIQ (Cotonou, Benin)	Tier 1 (Official Ministry)	Mid-cycle transition to formal deductive proof.

Ref ID	Curriculum & Level	Mathematical Domain / Competency	Verified Learning Expectation / Prescribed Content	Document Reference & Publishing Institution	Source Reliability	Research Notes / Status
SRC-07	Benin 3ème	Thalès, trigonométrie, réels, analytique	Théorème de Thalès; trigo dans le triangle rectangle; **R** et radicaux; calcul littéral, factorisation, équations-produits; géométrie analytique plane	*Guide pédagogique de mathématiques 3ème (NPE)*, MESTFP / DIPIQ (Cotonou, Benin)	Tier 1 (Official Ministry)	Statutory standard for BEPC examination.
SRC-08	Benin BEPC	Assessment Framework	Three-problem contextualised examination evaluated via C_a (Analyser), C_m (Mathématiser), C_o (Opérer), CP (Perfectionnement)	*Corrigés types et grilles d'évaluation officielles du BEPC Mathématiques*, Direction des Examens et Concours (DEC), MESTFP Bénin (Sessions 2020–2026)	Tier 2 (Official Examination)	Empirical verification of national examination rubrics.
SRC-09	Benin 2nde C/D	Fonctions, vecteurs, produit scalaire, second degré	Ensembles de nombres, fonctions de référence, trinôme du second degré (Δ), produit scalaire, Al-Kashi, trigo	*Programme d'études et Guide pédagogique de mathématiques 2nde C-D*, MESTFP / DIPIQ; *CIAM 2nde SM/SE*, EDICEF	Tier 1 / Tier 3	Foundation for high school scientific stream.
SRC-10	Benin 1ère C/D	Barycentres, dérivées, suites, probabilités	Barycentre de 2, 3, 4 points; dérivation et étude de fonctions; suites arithmétiques/géométriques; trigonométrie; combinatoire et probabilités	*Programme de Mathématiques de la classe de Première C/D (APC)*, MESTFP; *CIAM 1ère SM/SE*, EDICEF	Tier 1 / Tier 3	Advanced preparation for Baccalauréat.
SRC-11	Benin Baccalauréat	Examination Standards (Séries C et D)	Written 4-hour examination: analytical problem, spatial geometry/transformations, probability/statistics	*Annales et épreuves officielles du Baccalauréat Séries C et D*, Office du Baccalauréat du Bénin (Sessions 2018–2026)	Tier 2 (Official Examination)	Establishes terminal national expectations.

4. FEASIBILITY ANALYSIS OF IB–BENIN APC HARMONISATION

4.1 Structural and Epistemological Commonalities

1. Competency and Problem-Solving Orientation:

• Both programmes reject rote memorisation in favor of situated problem-solving. The Beninese *Situation d'Apprentissage* (SA) begins with a contextual challenge (*situation de départ*), mirroring the IB MYP requirement to explore mathematics through authentic *Global Contexts*.

2. Mathematical Progression:

• Both systems share an identical hierarchy of foundational concepts: from whole numbers to rational and real numbers; from synthetic planar geometry to coordinate geometry and vector algebra; from descriptive statistics to probability distributions; and from arithmetic sequences to functional calculus.

3. Assessment Criterion Correspondence:

• A direct isomorphism exists between the assessment dimensions of the two systems:

• IB **Criterion A (Knowing & understanding)** maps directly to Beninese **C_m (Mathématiser)** and **C_o (Opérer)**.

• IB **Criterion B (Investigating patterns)** shares deep cognitive alignment with the Beninese heuristic phase (*Recherche / Conjecturation* in SA).

• IB **Criterion C (Communicating)** corresponds to Beninese **Critère de Perfectionnement (CP)** and the formal requirement to justify deductive chains (*Raisonnement déductif structuré*).

- IB **Criterion D (Applying in real-life contexts)** maps directly to Beninese **C_a (Analyser)** within contextualised examination scenarios.

4.2 Structural and Methodological Divergences

1. Instructional Language and Technical Nomenclature:

- IB is delivered in English; the Beninese curriculum is delivered in French.
- Conceptual naming differences:
- *Médiatrice* vs *Perpendicular bisector*.
- *Théorème des milieux* vs *Midpoint theorem*.
- *Équation-produit* vs *Null-factor / zero-product property*.
- *Repère orthonormé* vs *Cartesian coordinate system / orthonormal frame*.
- *Suite arithmétique* vs *Arithmetic sequence* (French *suite* vs English *sequence*; English *series* refers specifically to the sum of a sequence).

2. Notational and Typographical Conventions:

- Decimal mark: Benin uses the comma (12,5); IB English uses the period (12.5).
- Thousand separator: Benin uses a thin non-breaking space (10 000); English often uses a comma (10,000) or thin space.
- Vector notation: Benin uses arrows (u■, AB■); IB permits bold roman (u) or vector arrows (v■ or overrightarrowAB).
- Interval notation: Benin uses reversed brackets for open intervals (]a, b[); IB permits both parenthesis notation ((a, b)) and bracket notation ([a, b[).
- Division symbol: Benin uses a : b or (a / b); IB English uses a ÷ b, a/b, or (a / b).
- Coordinates: Benin frequently separates coordinates with a semicolon (A(2 ; -3)) to avoid confusion with the decimal comma; IB English uses a comma (A(2, -3)).

3. Curricular Sequencing Mismatches:

- *Vectors*: Introduced synthetically and algebraically in Beninese 4ème (MYP 4) and 3ème (MYP 5); in standard IB MYP, formal vector operations are frequently delayed to extended MYP 5 or DP Topic 3.
- *Barycentres (Weighted Centers)*: Strongly emphasized in Beninese 1ère C/D as a major geometric tool for alignment and level curves; this topic is absent from the IB DP mathematics syllabus (which relies instead on vector equations, dot products, and linear combinations).
- *Radical Simplification and Surds*: Beninese 3ème expects high fluency in rationalising denominators containing conjugates ((a√(b) + c√(d))); IB Standard MYP addresses basic surds, with advanced rationalisation located in Extended MYP or DP.
- *Technology Integration*: IB DP mandates the use of Graphic Display Calculators (TI-84 Plus CE, TI-Nspire, or Casio fx-CG50) for numerical solver algorithms, regression, and statistical testing (Paper 2 & 3). In contrast, Beninese national examinations (BEPC and Baccalauréat) strictly restrict students to scientific, non-programmable calculators (or prohibit calculators entirely depending on local ministerial directives), necessitating sustained mental computation and algebraic paper-and-pencil mastery.

4.3 Conclusion on Harmonisation Feasibility

Pedagogical harmonisation is **feasible, advantageous, and mutually reinforcing**, provided that:

1. Oak does not compromise either syllabus: every required national competency must be fully secured, and every IB conceptual objective and inquiry criterion must be delivered.
2. A dual-track pedagogical strategy is maintained: shared core mathematical concepts are introduced together through deep conceptual understanding, followed by targeted, bilingual reinforcement of programme-specific examination protocols.
3. Bridge modules are introduced to reconcile sequencing mismatches (e.g., vectors, barycentres, formal surd arithmetic, and GDC technology).

End of Deliverable 1 — Research Report.